

Baud Rate

$$N = \frac{f_{BRCLK}}{\text{baud rate}} = \frac{32768}{9600} = 3.4133$$

$$\text{duration per bit sent or received (s)} = \frac{1}{\text{bps}}$$

ADC Conversion

$$\text{voltage divider} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

$$\text{adc conversion: } \frac{\text{pin volt}}{\text{max adc}} = \frac{\text{input voltage}}{\text{output voltage}}$$

$$\text{example: } \frac{3.3V}{4095 \text{ ADC}} = \frac{\text{Input Voltage}}{\text{Output ADC}}$$

$$\text{max adc} = 2^{\text{bit resolution}} - 1$$

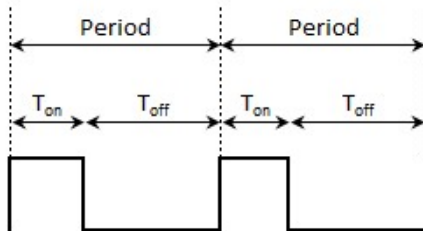
Steps for IC ADC Readings

1. Find min, max, and midpoint of g and v as x and y respectively
2. Create linear regression $y = mx + b$, sub in $v = mg + b$
3. Sub in g to get V then convert to ADC using adc conversion formula

Queue Testing

1. dequeue grabs oldest
2. test grabs the var right before
3. do math with the ascii values ig?

Duty Cycle



$$\text{Period} = 1 / \text{Frequency}$$

$$\text{Period} = T_{on} + T_{off}$$

$$\text{Duty Cycle} = \frac{T_{on}}{(T_{on} + T_{off})} \times 100$$

(On Percentage)

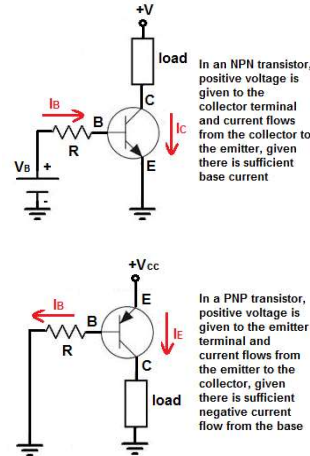
Duty Cycle Cont'd

$$TA1CCR0 = \left(1 \text{ MHz} \times \frac{1}{\text{freq}}\right) - 1$$

$$TA1CCR1 = n \text{ percent} \times TA1CCR0$$

$$\text{Duty Cycle (\%)} = \frac{TA1CCR1}{TA1CCR0} \times 100$$

Transistor Logic



Integrated Temperature Sensor

$$\text{Temp} (^{\circ}\text{C}) = (\text{adc raw} - \text{NADCL}) \times \left(\frac{TH - TL}{\text{NADCH} - \text{NADCL}} \right) + 30$$

$$^{\circ}\text{C} = \frac{5}{9} F - 32$$

$$^{\circ}\text{F} = \frac{9}{5} C + 32$$

Nyquist Frequency

$$f_o = 2f_{\text{sample}}$$

Raw Voltage Calibration

$$V_{\text{true}} = C_2 \times V_{\text{calculated-raw}} + C_1$$

$$\text{Percent Error} = (\text{Calculated} - \text{True}) / \text{True}$$

Parity Bit Calculations

1. Given 7E1 or 8E1 whatever
2. 7/8 = Number of Data bits
3. E = Even Parity (O for Odd)
4. Calculate data byte to binary
5. If more bits than limit, discard MSB
6. Count the number of 1's
7. Determine the parity bit using the table

Parity Type	If number of 1's is even	If number of 1's is odd
Even (E)	Parity bit = 0	Parity bit = 1
Odd (O)	Parity bit = 1	Parity bit = 0

If 7N1, No Parity bit, 1 Start Bit (0), 7 data bits, 1 Stop Bit (1)

Pulse Width Modulation

```
#include <msp430.h>
int main(void) {
    WDTCTL = WDTPW | WDTHOLD; // hold the watchdog timer
    PM5CTL0 &= ~LOCKLPM5; // clear LOCKLPM5 bit
    P1DIR |= BIT0; // output direction (P1.0)
    P1DIR |= BIT2; // output direction (P1.2)
    TA1CCR0 = 2000; // TA1CCR0 value
    TA1CCR1 = 500; // TA1CCR1 value
    TA1CTL = TASSEL_2 | MC_1 | TACLK; // TA1CTL setup
    TA1CCTL0 = CCIE; // enable CCIE
    TA1CCTL1 = CCIE; // enable CCIE
    __enable_interrupt(); // enable general interrupt
    while(1) {
        P1OUT ^= BIT0; // toggle (P1.0)
        __delay_cycles(250000); // delay
    }
    return 0;
}

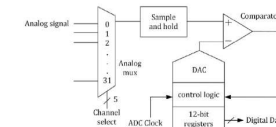
#pragma vector = TIMER1_A0_VECTOR
__interrupt void Timer1_A0_ISR(void) {
    P1OUT |= BIT2; // set (P1.2)
}

#pragma vector = TIMER1_A1_VECTOR
__interrupt void Timer1_A1_ISR(void) {
    if((TA1CCTL1 & CCIFG) != 0) { // check CCIFG flag
        P1OUT &= ~BIT2; // clear (P1.2)
        TA1CCTL1 &= ~CCIFG; // clear CCIFG flag
    }
}
```

Program 13.1. S/W PWM program using CCIFG flag (Interrupt).

ADC Converters

A conceptual block diagram of a 12-bit SAR with 32 input channels



Simplified functional block diagram of ADC12

